

# BASES ADMITTING SIX- AND EIGHT-DIGIT KAPREKAR FIXED POINTS: A COMPLETE CONGRUENCE CLASSIFICATION

AYDIN KAYA

**ABSTRACT.** For the Kaprekar transformation  $K_b$  (sort the base- $b$  digits in descending and ascending order and subtract) acting on integers with a fixed digit length  $n$ , we determine, for  $n = 6$  and  $n = 8$ , exactly which bases  $b \geq 2$  admit a fixed point. For  $n = 6$ : a fixed point exists if and only if  $2 \mid b$ , or  $b \equiv 6 \pmod{7}$ , or  $b \equiv 8 \pmod{9}$ , or  $b \equiv 10 \pmod{15}$ . For  $n = 8$ : if and only if  $3 \mid b$ , or  $b \equiv 10, 12 \pmod{15}$ , or  $b \equiv 11, 14 \pmod{17}$ , or  $b \equiv 13 \pmod{21}$ , or  $b \in \{2, 4, 8\}$ . All fixed points are given by explicit families, linear in  $b$ , together with a finite exceptional set confined to  $b \leq 10$ . The proofs are computer-assisted but exact: a channel decomposition of the borrow chain reduces the problem to finitely many linear systems over  $\mathbb{Q}$ , classified with exact rational arithmetic, and the indeterminate cases are closed by exact two-variable linear programming. We describe the structural transition between the two classifications: the moduli-doubling phenomenon for odd bases persists, while the universal even-base witness of  $n = 6$  is replaced at  $n = 8$  by a parity-blind  $3 \mid b$  family. A ground-truth scan for  $n = 10$  shows the same parity-blind  $3 \mid b$  family recurring and leaves the general- $n$  law open.

## 1. INTRODUCTION

Fix a base  $b \geq 2$  and a digit length  $n$ . For an integer  $x$  with exactly  $n$  digits in base  $b$  (leading digit nonzero), let  $\text{desc}(x)$  and  $\text{asc}(x)$  denote the integers obtained by sorting the digit string of  $x$  (leading zeros permitted after sorting) in descending and ascending order, and set

$$K_{b,n}(x) = \text{desc}(x) - \text{asc}(x).$$

Kaprekar’s celebrated observation [1] is that in base 10 with  $n = 4$  every non-repdigit integer reaches the fixed point 6174. The literature on the Kaprekar transformation splits into two natural axes. Along the first axis one fixes the *base* and studies all digit lengths: this is the programme completed for base 10 constants by Prichett, Ludington and Lapenta [6], refined by Iwasaki [9], and developed for general odd and even bases in the two-part work of Kay and Downes-Ward [7, 8]; Ludington [5] showed that every base admits only finitely many Kaprekar constants. Along the second axis one fixes the *digit length*  $n$  and varies the base. Here two questions must be distinguished. The classical, stronger question asks for *Kaprekar constants*: a unique fixed point attracting every non-repdigit  $n$ -digit integer. Prichett and Hasse [2] determined all bases with a four-digit Kaprekar constant; Lapenta, Ludington and Prichett [3] gave an algorithm for self-producing (= fixed)  $r$ -digit  $g$ -adic integers. Thakur [4] settled the odd lengths  $n = 5, 7, 9$  (with a single base each for  $n = 7, 9$ ), gave partial data for  $n = 6, 8$  (Kaprekar constants found only at  $b = 13, 17$  for  $n = 6$  among  $b \leq 39$ , and only at  $b = 3$  for  $n = 8$  among  $b \leq 17$ ), and posed as an open problem exactly the existence question we answer here (“characterization of  $(B, D)$  such that  $\text{FP}(B, D)$  is non-empty”). The very recent cross-base study of  $n = 4$  dynamics in [10] shows the axis remains active. The weaker *existence* question

---

2020 *Mathematics Subject Classification.* 11A63 (primary), 05A17, 11B50 (secondary).

*Key words and phrases.* Kaprekar transformation, fixed points, digit maps, radix representations, computer-assisted proof.

(which bases admit *any*  $n$ -digit fixed point) is the natural first layer of that programme, and it is the question we settle here for the even lengths  $n = 6$  and  $n = 8$ . The distinction matters: for instance, our scanner shows that for  $b \leq 60$  five-digit fixed points exist precisely for  $b = 2$  and  $3 \mid b$ , whereas five-digit Kaprekar *constants* are far rarer.

To our knowledge the even lengths  $n \geq 6$  have remained open on the second axis. This paper closes the first two cases.

**Theorem 1** ( $n = 6$ ). *A base  $b \geq 2$  admits a six-digit Kaprekar fixed point if and only if*

$$2 \mid b \quad \vee \quad b \equiv 6 \pmod{7} \quad \vee \quad b \equiv 8 \pmod{9} \quad \vee \quad b \equiv 10 \pmod{15}.$$

*All fixed points belong to the four explicit families of Table 1, apart from a finite exceptional set of points confined to bases  $b \leq 10$  (each such base already being covered by the families). Restricted to odd bases the condition becomes  $b \equiv 13 \pmod{14} \vee b \equiv 17 \pmod{18} \vee b \equiv 25 \pmod{30}$ .*

**Theorem 2** ( $n = 8$ ). *A base  $b \geq 2$  admits an eight-digit Kaprekar fixed point if and only if*

$$3 \mid b \vee b \equiv 10, 12 \pmod{15} \vee b \equiv 11, 14 \pmod{17} \vee b \equiv 13 \pmod{21} \vee b \in \{2, 4, 8\}.$$

*All fixed points belong to the six explicit families of Table 2, apart from nine exceptional points confined to  $b \in \{2, 4, 6, 8, 10\}$ ; the bases 2, 4, 8 are the only ones whose fixed points are all exceptional. Restricted to odd bases the condition becomes  $b \equiv 3 \pmod{6} \vee b \equiv 25, 27 \pmod{30} \vee b \equiv 11, 31 \pmod{34} \vee b \equiv 13 \pmod{42}$ .*

The proofs are computer-assisted in the strong sense: every step is either an elementary identity (the borrow-chain successor formulas of Section 2) or a finite computation carried out in exact rational arithmetic, with the two logical directions treated asymmetrically: every *existence* claim is sealed by directly verifying  $K_{b,n}(x) = x$  for witnesses, while every *emptiness* claim uses only necessary linear constraints (Section 3). The complete classification data and the verification code accompany the paper as ancillary files.

Structurally, the two theorems exhibit a striking transition (Section 6): the moduli-doubling phenomenon for odd bases (each family modulus  $D$  doubles to  $2D$  under the odd-base restriction) persists from  $n = 6$  to  $n = 8$ , but the palindromic witness that makes *every* even base admissible at  $n = 6$  has no analogue at  $n = 8$ ; its structural slot is occupied by a parity-blind  $3 \mid b$  family, so that infinitely many even bases (e.g.  $b \equiv 16 \pmod{30}$  outside the other families) admit no eight-digit fixed point. The asymptotic density of admissible bases is  $68/105 \approx 0.648$  for  $n = 6$  and  $60/119 \approx 0.504$  for  $n = 8$ , the latter being identical for odd and even bases.

## 2. PRELIMINARIES AND THE CHANNEL DECOMPOSITION

Throughout,  $n$  is even,  $h := n/2$ , and  $B := b - 1$ .

**Definition 3.** Let  $x$  have base- $b$  digits sorted as  $a_0 \geq a_1 \geq \dots \geq a_{n-1}$ . The *difference vector* of  $x$  is  $d(x) = (d_0, \dots, d_{h-1})$  with  $d_i = a_i - a_{n-1-i}$ .

Sortedness forces  $d_0 \geq d_1 \geq \dots \geq d_{h-1} \geq 0$ . A direct computation gives

$$(1) \quad K_{b,n}(x) = \sum_{i=0}^{h-1} d_i (b^{n-1-i} - b^i),$$

so  $K_{b,n}$  depends on  $x$  only through  $d(x)$ , and  $x$  is a fixed point if and only if  $x$  equals the right-hand side of (1) evaluated at  $d(x)$ . This identity also underlies our ground-truth scanner: enumerating difference vectors costs  $O(b^h)$  per base instead of  $O(b^n)$ .

Resolving the borrow chain in (1) yields the following *channel decomposition*. Let  $t \in \{0, \dots, h-1\}$  be the number of trailing zeros of  $d(x)$  and  $m := h - t$ .

**Lemma 4** (successor formulas). *If  $x$  is a non-repdigit  $n$ -digit integer with difference vector  $d$  having  $t$  trailing zeros, then the digit string of  $K_{b,n}(x)$ , from most to least significant, is*

$$[d_0, d_1, \dots, d_{m-2}, d_{m-1} - 1, \underbrace{B, \dots, B}_{2t}, B - d_{m-1}, \dots, B - d_1, B + 1 - d_0].$$

*In particular the digit sum of any  $n$ -digit fixed point equals  $(h+t)B$ , and the digit-sum multiples  $kB$  with  $k = h, \dots, 2h-1$  partition all fixed points into  $h$  channels.*

*Proof.* The column differences of desc-asc, from most to least significant, are  $d_0, \dots, d_{h-1}, -d_{h-1}, \dots, -d_0$ . Working from the right: the column  $-d_0$  borrows (as  $d_0 \geq 1$ ), producing digit  $b - d_0$ ; each subsequent column  $-d_i - 1$  with  $i \geq 1$  is negative, producing  $B - d_i$ ; the  $2t$  central columns  $0 - 1$  propagate the borrow producing digit  $B$ ; the first nonzero column from the centre,  $d_{m-1} - 1 \geq 0$ , absorbs the borrow; the remaining columns are unchanged. Summing the resulting digits gives  $(h+t)B$ .  $\square$

For  $n = 6$  the channels have digit sums  $3B, 4B, 5B$ ; for  $n = 8$  they are  $4B, \dots, 7B$ . Empirically (and, after Section 3, provably) only the channels  $t = 0$  and  $t = 1$  carry infinite families; channels  $t \geq 2$  are overdetermined by  $t$  consistency conditions and collapse to finitely many small-base points.

### 3. METHOD: CELLS, EXACT CLASSIFICATION, AND CLOSURE

Fix  $n$  and a channel  $t$ , and let  $E_1, \dots, E_n$  denote the  $n$  successor expressions of Lemma 4, each linear in  $(d_0, \dots, d_{m-1})$  and  $B$ . If  $x$  is a fixed point, its multiset of digits is  $\{E_j\}$ ; choosing a permutation  $\sigma$  that sorts the expressions into weakly decreasing order therefore places  $x$  in the cell  $\sigma$ , where the fixed-point condition becomes the linear system

$$E_{\sigma(i)} - E_{\sigma(n-1-i)} = d_i \quad (i = 0, \dots, h-1, \text{ with } d_i := 0 \text{ for } i \geq m),$$

that is,  $A_\sigma d = B c_\sigma + e_\sigma$  with integer data. Here  $A_\sigma$  is the  $h \times m$  integer matrix whose  $i$ -th row lists the coefficients of the unknowns  $(d_0, \dots, d_{m-1})$  in the left-hand expression  $E_{\sigma(i)} - E_{\sigma(n-1-i)} - d_i$ , while  $c_\sigma$  and  $e_\sigma$  in  $\mathbb{Z}^h$  collect, row by row, the coefficient of  $B$  and the constant term on the right. Every fixed point lies in at least one cell (ties are broken arbitrarily; all cell inequalities are closed), so classifying all cells classifies all fixed points.

Each cell is classified over  $\mathbb{Q}$  by row reduction in exact rational arithmetic. Writing  $\Delta := \det A_\sigma$  when the system is square and of full rank, a unique solution then has the closed form  $d_i = (\alpha_i B + \beta_i)/\Delta$ . The cells fall into four classes: (a) a unique solution valid for all  $B$ ; (b) a unique solution only at a single value of  $B$ ; (c) inconsistent (no solution); and (d) underdetermined (a positive-dimensional solution set). For cells in class (a) we then impose the full validity system — the sorting inequalities of  $\sigma$ , the digit bounds  $0 \leq E_j \leq B$ ,  $d_0 \geq 1$ , the channel condition  $d_{m-1} \geq 1$ , and monotonicity of  $d$  — all of which are linear in  $B$ , giving exact thresholds; integrality of the  $d_i$  imposes congruences on  $B$  modulo  $\Delta$ . A cell whose inequalities hold for all large  $B$  and whose congruences are solvable yields an *infinite family*; a cell whose inequalities hold only in a window  $[B_{\min}, B_{\max}]$  is exhausted by direct enumeration.

**Worked examples** ( $n = 6$ ). To make the four classes concrete, take  $n = 6$ , where the six successor expressions are  $E = (d_0, d_1, d_2 - 1, B - d_2, B - d_1, b - d_0)$  in channel  $t = 0$  (three unknowns) and  $E = (d_0, d_1 - 1, B, B, B - d_1, b - d_0)$  in channel  $t = 1$  (two unknowns). One representative of each class:

(a) *Unique for all  $B$  — an infinite family.* In channel  $t = 0$ , the ordering  $[B - d_2, d_0, d_1, B - d_1, b - d_0, d_2 - 1]$  yields

$$-d_0 - 2d_2 = -B - 1, \quad 2d_0 - d_1 = B + 1, \quad 2d_1 - d_2 = B,$$

with the unique solution  $d = (\frac{7B+5}{9}, \frac{5B+1}{9}, \frac{B+2}{9})$ , integral and valid for all  $b \geq 8$ . This is precisely the  $b \equiv 8 \pmod{9}$  family of Table 1, with  $\Delta = 9$ .

(b) *Unique at a single B*. In channel  $t = 1$ , the ordering  $[d_0, B, B, b - d_0, d_1 - 1, B - d_1]$  yields  $d_1 = B$ ,  $2d_1 = B + 1$ , and  $d_0 = 1$ ; the first two force  $B = 1$ , so this cell contributes a fixed point only at the single base  $b = 2$ .

(c) *Inconsistent*. In channel  $t = 0$ , the ordering  $[d_0, d_2 - 1, d_1, b - d_0, B - d_1, B - d_2]$  yields both  $d_2 = B$  and  $d_2 = B + 1$ ; row reduction leaves the residual equation  $0 = -1$ , so the cell is empty for every base.

(d) *Underdetermined*. In channel  $t = 0$ , the ordering  $[d_0, d_2 - 1, d_1, B - d_2, b - d_0, B - d_1]$  yields  $d_1 = B$  (twice) together with  $d_0 - d_1 + d_2 = B + 2$ ; the coefficient matrix has rank  $2 < 3$ , leaving one free parameter. Such cells are closed by the linear program of principle (ii) below.

Two design principles keep the argument airtight.

(i) *One-way door*. Every existence claim (family members, exceptional points) is finally verified by evaluating (1) and checking  $K_{b,n}(x) = x$  directly; every emptiness claim uses only constraints that are provably necessary for a fixed point, all closed under equality. An off-by-one in a constraint can therefore cost efficiency, never correctness.

(ii) *Exact closure of underdetermined cells*. In an underdetermined cell the solution set is affine in  $(B, u)$ , where the free parameter  $u$  is itself one of the  $d_j \in [0, B]$ . In every such cell arising for  $n \in \{6, 8\}$  the free dimension is 1, so the validity region is a polyhedron in the  $(B, u)$ -plane. Since the constraints include  $B \geq 1$  and  $0 \leq u \leq B$ , the region contains no line; hence, if nonempty, it has a vertex, and vertex enumeration over all constraint pairs (again in exact arithmetic) decides emptiness and, when bounded, computes  $\max B$  exactly. Cells fixing a single  $B$  (the “finite- $B$ ” underdetermined cells) are exhausted by direct enumeration of the free parameters at that base.

Finally, the entire pipeline is anchored to ground truth: an exhaustive difference-vector scan of all bases  $b \leq 60$  (and, as an independent convention check, a naive full-range scan over all  $n$ -digit integers for small bases), against which every symbolic conclusion is compared point-by-point, plus an extension scan ( $b \leq 150$  for  $n = 6$ ,  $b \leq 100$  for  $n = 8$ ) confirming the classifications beyond the range used to build them. A further audit re-derives the verdicts of 1200 randomly sampled cells per length by exhaustive enumeration of the descending  $d$ -cone at two bases, independently of the linear algebra code.

#### 4. THE CLASSIFICATION FOR $n = 6$

TABLE 1. The four families for  $n = 6$  ( $B = b - 1$ ; missing  $d$ -components are 0).

| Channel | Congruence              | $d$ -vector                                           | First members   |
|---------|-------------------------|-------------------------------------------------------|-----------------|
| $t = 1$ | $b \equiv 0 \pmod{2}$   | $(\frac{B+1}{2}, \frac{B+1}{2})$ (palindromic $M_b$ ) | 2, 4, 6, ...    |
| $t = 0$ | $b \equiv 6 \pmod{7}$   | $(\frac{5B+3}{7}, \frac{3B-1}{7}, \frac{B+2}{7})$     | 6, 13, 20, ...  |
| $t = 0$ | $b \equiv 8 \pmod{9}$   | $(\frac{7B+5}{9}, \frac{5B+1}{9}, \frac{B+2}{9})$     | 8, 17, 26, ...  |
| $t = 0$ | $b \equiv 10 \pmod{15}$ | $(\frac{3B+3}{5}, \frac{B}{3}, \frac{B+1}{5})$        | 10, 25, 40, ... |

The cell census for the main channel  $t = 0$  of  $n = 6$ : of the 720 orderings, 568 have a unique solution for all  $B$  (of which exactly 3 survive as infinite families), 140 admit no valid base, and 12 are underdetermined. The twelve underdetermined cells are closed exactly by the linear programming argument of Section 3: eleven have empty validity regions and one is bounded with  $\max b \leq 60$ , inside the exhaustively scanned range. The channel  $t = 1$  contains no underdetermined cells and exactly one family, with modulus 2 and residue 0; consequently *no odd base has a fixed point in the  $4B$  channel*, previously an empirical observation and now

a consequence of the classification. Ground truth (52 fixed points for  $b \leq 60$ , digit-sum split  $k = 3:22$ ,  $k = 4:30$ ) matches the families plus exceptional points exactly, as does the extension scan to  $b \leq 150$ .

The density of admissible bases is  $408/630 = 68/105$ ; among odd bases it is  $186/630 = 31/105 \approx 0.295$ , matching the classical count.

## 5. THE CLASSIFICATION FOR $n = 8$

TABLE 2. The six families for  $n = 8$ .

| Channel | Congruence              | $d$ -vector                                                                        | First members    |
|---------|-------------------------|------------------------------------------------------------------------------------|------------------|
| $t = 1$ | $b \equiv 0 \pmod{3}$   | $\left(\frac{2B+2}{3}, \frac{2B-1}{3}, \frac{B+1}{3}\right)$                       | $3, 6, 9, \dots$ |
| $t = 0$ | $b \equiv 10 \pmod{15}$ | $\left(\frac{3B+3}{5}, \frac{B}{3}, \frac{B}{3}, \frac{B+1}{3}\right)$             | $10, 25, 40, 55$ |
| $t = 0$ | $b \equiv 12 \pmod{15}$ | $\left(\frac{13B+7}{15}, \frac{11B-1}{15}, \frac{7B-2}{15}, \frac{B+4}{15}\right)$ | $12, 27, 42, 57$ |
| $t = 0$ | $b \equiv 11 \pmod{17}$ | $\left(\frac{11B+9}{17}, \frac{7B-2}{17}, \frac{5B+1}{17}, \frac{3B+4}{17}\right)$ | $11, 28, 45$     |
| $t = 0$ | $b \equiv 14 \pmod{17}$ | $\left(\frac{15B+9}{17}, \frac{13B+1}{17}, \frac{9B+2}{17}, \frac{B+4}{17}\right)$ | $14, 31, 48$     |
| $t = 0$ | $b \equiv 13 \pmod{21}$ | $\left(\frac{5B+3}{7}, \frac{3B-1}{7}, \frac{B}{3}, \frac{B+2}{7}\right)$          | $13, 34, 55$     |

The nine exceptional points have difference vectors  $(1, 1, 1, 1)$ ,  $(1, 1, 1, 0)$ ,  $(1, 1, 0, 0)$  at  $b = 2$ ;  $(3, 1, 1, 1)$ ,  $(3, 3, 1, 1)$ ,  $(3, 3, 3, 1)$  at  $b = 4$ ;  $(5, 5, 3, 1)$  at  $b = 6$ ;  $(7, 5, 3, 1)$  at  $b = 8$ ; and  $(9, 7, 5, 1)$  at  $b = 10$ .

Cell census: channel  $t = 0$  has 40320 cells (33344 unique-for-all- $B$ , of which 9 live cells give the five families of Table 2 after deduplication; 505 bounded windows, all exhausted; 5152 inconsistent; 1824 underdetermined). Channel  $t = 1$  has 20160 distinct cells and exactly one family; channels  $t = 2, 3$  close completely (only the  $b = 2$  point, resp. nothing). Of the underdetermined cells, all 1162 finite- $B$  ones fix bases  $b \leq 60$ , covered by the exhaustive scan; the 824 all- $B$  ones all have free dimension one, and the exact LP closure finds 779 empty regions and 45 bounded with  $\max b \leq 4$ , and none unbounded. Hence the classification is complete for every base.

**Remark 5** (base ten). The two classical eight-digit fixed points of base 10 (listed, among others, in OEIS [11]) are of different kinds: 63317664 (difference vector  $(6, 3, 3, 2)$ ) is the first member of the  $b \equiv 10 \pmod{15}$  family, while 97508421  $((9, 7, 5, 1))$  is exceptional: it belongs to no infinite family.

The density of admissible bases is  $900/1785 = 60/119 \approx 0.504$ ; because every family modulus is odd, the density is the same among odd and among even bases.

## 6. STRUCTURAL COMPARISON AND THE PARITY MECHANISM

As an independent check, Thakur's data [4] on which bases carry a full attracting Kaprekar constant (a strictly stronger property than mere existence of a fixed point) sit inside our classification, as they must: his  $n = 6$  constants at  $b = 13, 17$  are the smallest members of our  $b \equiv 13 \pmod{14}$  and  $b \equiv 17 \pmod{18}$  families, and his  $n = 8$  constant at  $b = 3$  is the smallest member of our  $3 \mid b$  family.

This table has two sources. In channel  $t$  the system has  $h$  equations and  $m = h - t$  unknowns, so  $t = 0$  is square (many families),  $t = 1$  carries exactly one consistency condition (one distinguished family), and  $t \geq 2$  is overdetermined (finite collapse). Separately, the divisor of the single  $t = 1$  family is even only at  $n = 6$ : it is 2 there, forcing  $2 \mid b$  and creating the even/odd asymmetry, but 3 at  $n = 8$  (and again at  $n = 10$ ; see Remark 6), which is parity-blind. Since all  $t = 0$  moduli

TABLE 3. Structural comparison.

|                         | $n = 6$                     | $n = 8$                              |
|-------------------------|-----------------------------|--------------------------------------|
| channels (live)         | $3B, 4B$ (of $3B, 4B, 5B$ ) | $4B, 5B$ (of $4B, \dots, 7B$ )       |
| even bases              | all admissible              | not all ( $16, 20, 22, \dots$ empty) |
| $t = 1$ witness divisor | 2 ( $M_b$ , palindromic)    | 3                                    |
| families / moduli       | 4 / $\{2, 7, 9, 15\}$       | 6 / $\{3, 15, 15, 17, 17, 21\}$      |
| odd-base moduli         | 14, 18, 30 (doubled)        | 6, 30, 34, 42 (doubled)              |
| density (all / odd)     | 68/105 / 31/105             | 60/119 / 60/119                      |

are odd, the parity wall of  $n = 6$  appears special to that divisor being 2 rather than a general feature; what fixes the  $t = 1$  divisor for general  $n$  we leave open.

**Remark 6** ( $n = 10$ ). A ground-truth scan of all bases  $b \leq 45$  (ancillary file `check_n10.py`) fixes the qualitative picture and corrects a tempting guess. Of the five channels ( $5B, \dots, 9B$ ) only  $t = 0$  and  $t = 1$  carry infinite families, while  $t = 2, 3$  collapse to  $b = 2$ . The distinguished  $t = 1$  family is

$$b \equiv 0 \pmod{3}, \quad d = \left( \frac{2B+2}{3}, \frac{2B+2}{3}, \frac{B+1}{3}, \frac{B+1}{3}, 0 \right),$$

of divisor 3 — *not* the value  $h - 1 = 4$  that the  $n = 6, 8$  divisors  $(2, 3)$  might suggest. There is thus no return of an even-base advantage;  $3 \mid b$  persists, parity-blind, as at  $n = 8$ . The  $t = 0$  families extend those of Table 2 structurally: for example  $b \equiv 10 \pmod{15}$  recurs as  $d = \left( \frac{3B+3}{5}, \frac{B}{3}, \frac{B}{3}, \frac{B}{3}, \frac{B+1}{5} \right)$ , the  $n = 8$  family of the same congruence with one additional  $\frac{B}{3}$  entry, and the  $t = 0$  moduli remain odd.

Two problems remain open. A complete  $n = 10$  classification is feasible once the  $10!$  cell enumeration is reduced to the riffle-shuffle interleavings of the two internally-ordered successor blocks (a few hundred per channel, versus  $10!$  naively). And the general- $n$  law governing the  $t = 1$  divisor is genuinely open: its first three values 2, 3, 3 do not follow the naive  $h - 1$ , a reminder that two data points are not a pattern.

## 7. REPRODUCIBILITY

All computations use exact integer and rational arithmetic (no floating point). The ancillary files contain: the core scanner and the general even- $n$  pipeline (`core.py`, `genpipe.py`, and the  $n = 8$  phase scripts), the ground-truth data ( $b \leq 60$ ), the complete cell classifications, and the audit scripts. A single command reruns each stage; the  $n = 6$  hardening run reproduces the historical cell census 568/140/12 found in the author’s original computation, an independent cross-validation of both implementations.

## ACKNOWLEDGEMENTS

This work was carried out in an extended human–AI collaboration. The classification pipeline, the symbolic derivations, and the drafts of the proofs and of this paper were developed jointly with Anthropic’s Claude models: the  $n = 6$  classification originally with Claude Opus, and the  $n = 8$  generalisation, the exact closure arguments, the audits, and this manuscript with Claude (Table 5). In line with arXiv policy, the AI system is not an author; the author directed the research programme, set the verification standards, and takes responsibility for the content. All mathematical claims were machine-verified in exact arithmetic and independently audited as described in Section 3.

## REFERENCES

- [1] D. R. Kaprekar, *An interesting property of the number 6174*, Scripta Math. **21** (1955), 304.
- [2] G. D. Prichett and H. Hasse, *The determination of all four-digit Kaprekar constants*, J. Reine Angew. Math. **299/300** (1978), 113–124.
- [3] J. F. Lapenta, A. L. Ludington, and G. D. Prichett, *An algorithm to determine self-producing  $r$ -digit  $g$ -adic integers*, J. Reine Angew. Math. **310** (1979), 100–110.
- [4] D. S. Thakur, *Kaprekar phenomena*, Proceedings of the Ropar Conference, RMS Lecture Notes Series No. 26 (2019), 1–10.
- [5] A. L. Ludington, *A bound on Kaprekar constants*, J. Reine Angew. Math. **310** (1979), 196–203.
- [6] G. D. Prichett, A. L. Ludington, and J. F. Lapenta, *The determination of all decadic Kaprekar constants*, Fibonacci Quart. **19** (1981), no. 1, 45–52.
- [7] A. Kay and K. Downes-Ward, *Fixed points and cycles of the Kaprekar transformation: 1. Odd bases*, J. Integer Seq. **25** (2022), Article 22.6.7.
- [8] A. Kay and K. Downes-Ward, *Fixed points and cycles of the Kaprekar transformation: 2. Even bases*, arXiv:2408.12257 (2024).
- [9] H. Iwasaki, *A new classification of the Kaprekar numbers*, Fibonacci Quart. **62** (2024), no. 4, 275–281.
- [10] E. Chen, K. Ono, R. E. Schwartz, and D. S. Thakur, *Four-digit Kaprekar dynamics in odd bases*, arXiv:2606.20439 (2026).
- [11] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A099009, <https://oeis.org/A099009>.

INDEPENDENT RESEARCHER

Email address: [kaya.ayd@gmail.com](mailto:kaya.ayd@gmail.com)